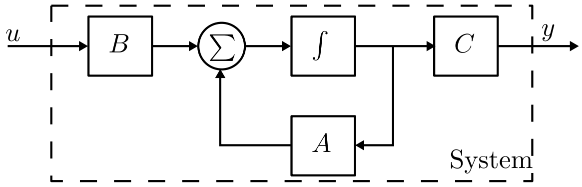
Test: Integral Control

1. Complete the following state space formulation, such that the integral state x_I satisfies $x_I(t) = \int_0^t (y(\tau) - r(\tau))d\tau$.

$$\begin{bmatrix} \dot{x} \\ \dot{x}_I \end{bmatrix} = \begin{bmatrix} A & 0 \\ & \end{bmatrix} \begin{bmatrix} x \\ x_I \end{bmatrix} + \begin{bmatrix} 0 \\ \end{bmatrix} r$$
$$y = \begin{bmatrix} C & 0 \end{bmatrix} \begin{bmatrix} x \\ x_I \end{bmatrix}$$

2. Complete the following diagram with a *full order observer* and a *state feedback* **with** *integral action.*



3. Determine if the following system is controllable and observable, by calculating its controllability matrix and observability matrix.

$$\dot{x} = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} x.$$

4. Write down an expression for the eigenvalues of the closed loop system given by

$$\text{System:} \quad \begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$
$$\text{Observer:} \quad \begin{cases} \dot{\hat{x}} = A\hat{x} + Bu + L(C\hat{x} - y) \\ y = C\hat{x} \end{cases}$$
$$\text{Controller:} \quad u = F\hat{x}.$$